

Python SQLite3 — Practice Assignment

Part A — Basic

1. Create a SQLite database named `college.db`.
2. Create a `students` table with:
 - `id`
 - `name`
 - `age`
 - `course`
 - `marks`
3. Insert 5 student records.
4. Display all students.
5. Display only student names.
6. Display students who scored more than 75 marks.
7. Find students belonging to the Python course.
8. Find the student with a particular ID.
9. Update the marks of one student.
10. Delete one student.

Part B — Intermediate

11. Insert 10 students using `executemany()`.
12. Display students in descending order of marks.
13. Display the top 3 students.
14. Find the total number of students using `COUNT()`.
15. Find the average marks using `AVG()`.
16. Find the highest and lowest marks.
17. Count students course-wise using `GROUP BY`.
18. Find the average marks for each course.
19. Search students whose names start with "A".
20. Search students whose marks are between 60 and 90.
21. Create a menu-driven CRUD application.
22. Add exception handling using `try-except`.
23. Use parameterized queries for all user inputs.
24. Implement `commit()` and `rollback()`.

Part C — Advanced

25. Create three tables: `students`, `courses`, and `enrollments`.
26. Establish relationships using primary and foreign keys.
27. Display student names along with their course names using `JOIN`.
28. Find students who are not enrolled in any course.
29. Find the number of students in each course.
30. Create a Python `Database` class containing CRUD methods.
31. Create a student search feature with multiple filters.
32. Add pagination using `LIMIT` and `OFFSET`.
33. Create indexes and investigate their effect on queries.
34. Use SQLite transactions to insert/update multiple records safely.

Final Challenge

Build a **Student Management System using Python + SQLite3** with:

1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Top Students
7. Course-wise Statistics
8. Average Marks
9. Student Count
10. Exit

Requirement: Use functions, SQLite3, parameterized queries, exception handling, `commit()`, `rollback()`, and a menu-driven interface.